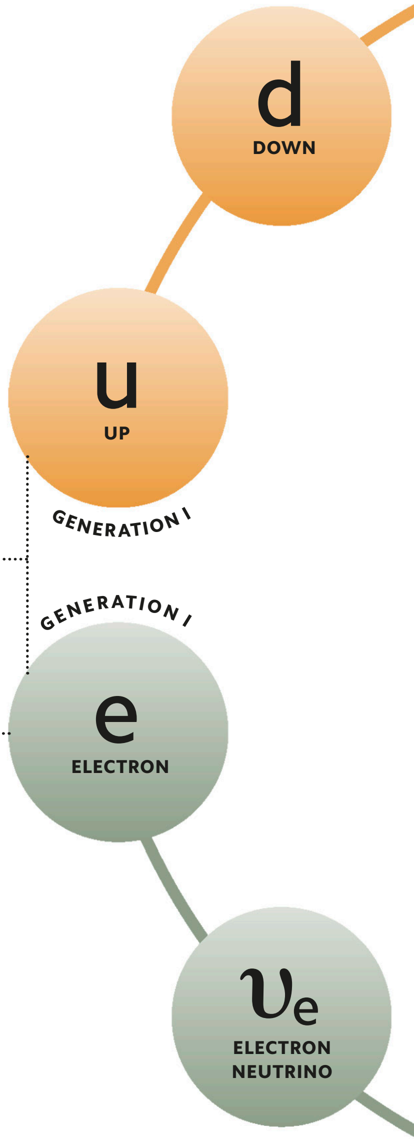
THE QUANTUM WORLD EXPLAINED

The Standard Model is the most successful theory for organizing the quantum world. It describes everything in terms of interactions played out by the set of elementary matter particles (fermions), force-carrying particles (gauge bosons), and the Higgs boson. Despite the success of the theory in predicting experimental results, it is considered a work in progress as there are some phenomena that cannot yet be explained (see pp.130–31).



CREATING MATTER

Leptons and quarks are fermions (with half-integer spin). They are the smallest building blocks of matter.

LEPTONS

Leptons come in three “generations” and two types: charged and neutral. The charged leptons are the electron, muon, and tau. The neutrinos are neutral.

> **”The Standard Model is so complex it would be hard to put it on a T-shirt—though not impossible; you'd just have to write kind of small.”**
>
> Steven Weinberg